Modularization
ESB Product Evaluation

Version 3.0

Jérémie Granat

Felix Winistörfer

15 March 2013

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

HP BSC Solution & Service Portfolio

HP Banking Service Center, Bern

Banking Expertise and Solutions

Infrastructure Outsourcing

- Consulting & Integration (plan & build)
- Operations (run)

Data Center Services (Hosting- and Infrastructure Mgmt)

Security, Compliance and Continuity Services

Workplace Services

Network Services

Application Services

- Consulting, Development & Integration (plan & build)
- Management & Operations (run)

Core Banking Applications and Peripheral Systems

Complementary Enterprise Applications

Business Process Outsourcing

- Consulting (plan & build)
- Management and Operations (run)

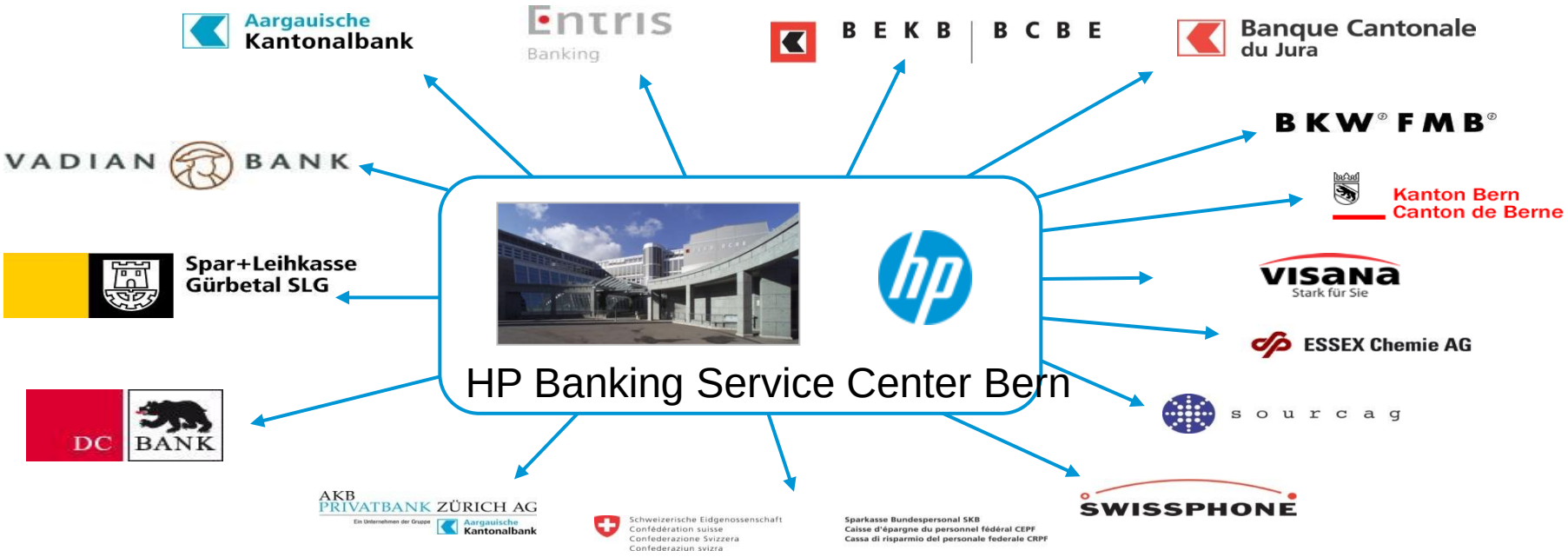
Output and Document Management

Other BPO Services

Swiss and Global Delivery | Swiss Data Centers

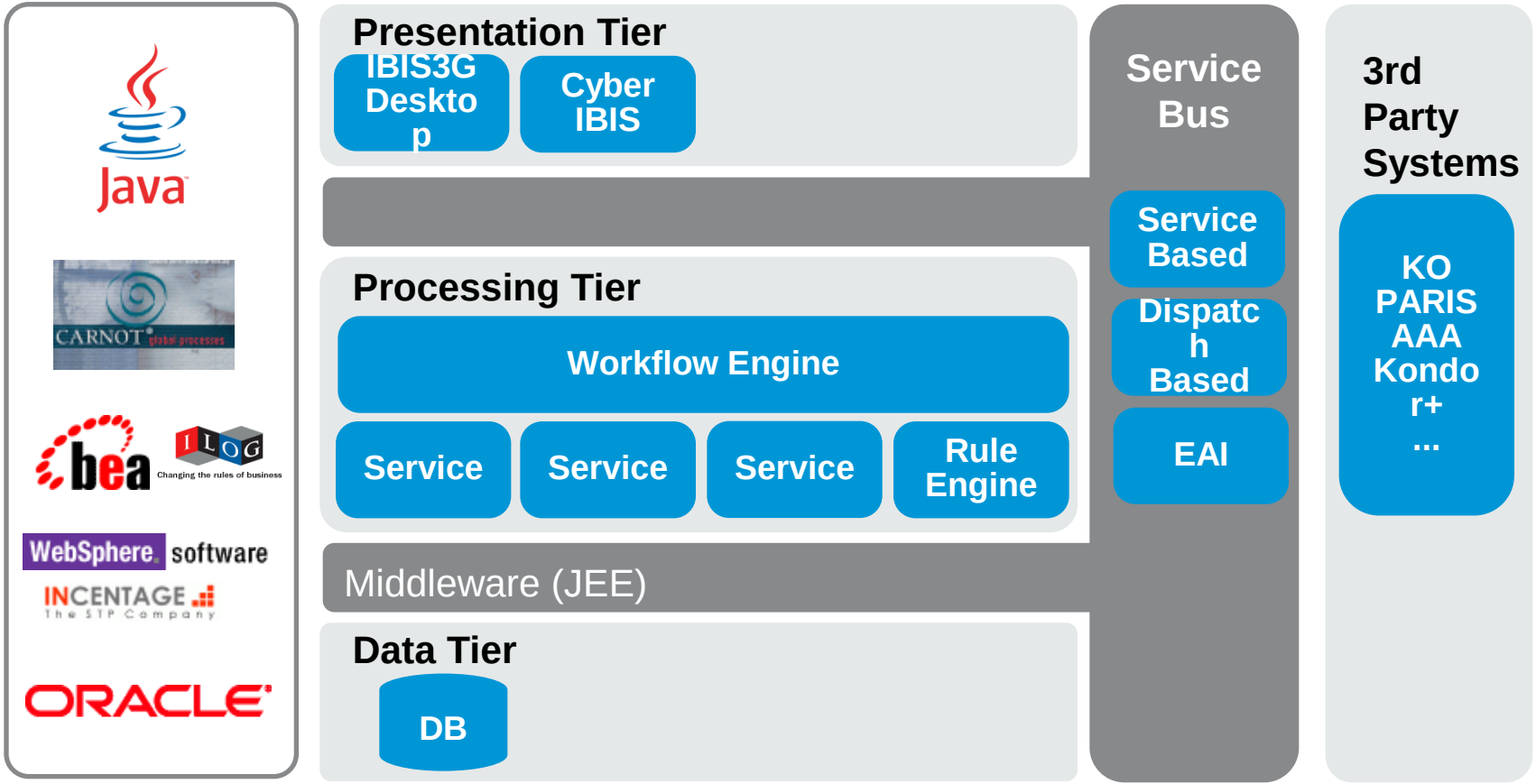
© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

HP Banking Service Center (BSC) Customers



© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

IBIS3G IT Architecture



© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Agenda

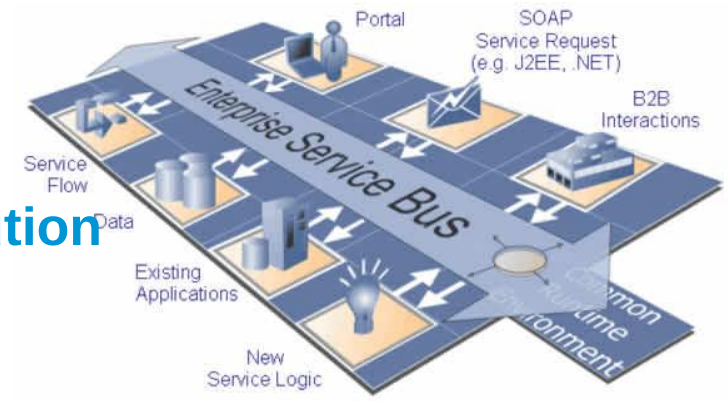
1. Goal of the project
2. Why Open Source Software
3. JMS Evaluation
4. ESB Evaluation
5. POC Results
6. Lessons Learned
7. Q&A

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Activity Goal
Find an Enterprise Service Bus to enable

-Modularisation of the different IBIS3G applications for better Go-To-Market

-Cost reduction in ongoing operation execution



© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Scope and basic requirements

Scope

-Basic Messaging (JMS)

-EJB / Sync Communication

-Basic Orchestration

Requirements

-Reliability

-High availability

-Professional support

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Community vs. Commercial Open Source Software vs. Commercial Software

Community OSS

Open Source direct from the community

Pros

-Newest Technologies

-Fast reaction to community needs

-Easy to use in development

Cons

- No guaranteed support
- Very fast lifecycle
- Bugs in releases
- Support can be stopped at any time

Commercial OSS

Open Source with Professional maintenance contract

Pros

- Guaranteed support (SOA-P: 10 years standard support)
- Hardened software releases
- “Normal” lifecycles

Cons

- Not on the “cutting edge”
- Problems if developer uses community edition

Commercial Software

Not Open Source

Pros

- Guaranteed support
- Hardened software releases
- “Normal” lifecycles

Cons

- Not on the “cutting edge”
- Ev. Vendor Lock-In
- Licenses

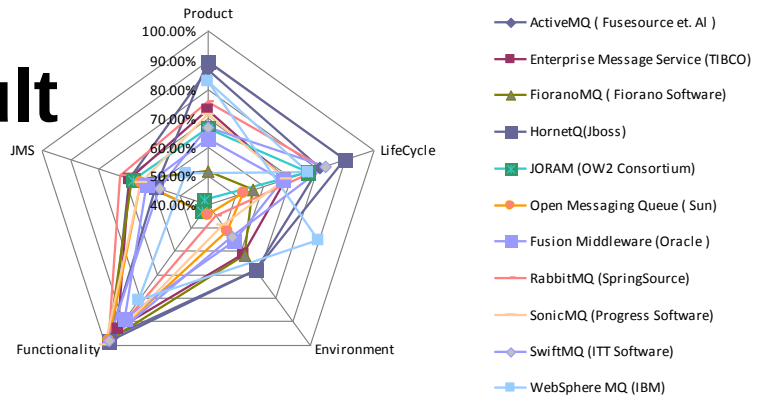
© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Evaluated JMS Solutions

PRODUCT	VENDOR	EDITION	LICENSE	NOTES
ActiveMQ	Apache	5.5.1	Apache Software Foundation License 2.0	Open Source. Commercial support offered by different vendors.
Enterprise Message Service	TIBCO	6.1	Commercial	RTView Monitoring software for EMS Systems
FioranoMQ	Fiorano Software	9.4.0	Commercial	JMS Performance Comparison
HornetQ	JBoss	2.2.5	Apache Software Foundation License 2.0	HornetQ is part of the product EAP 5.1.1 commercially supported by RedHat
JORAM	OW2 Consortium	5.7.0	GNU LGPL	Professional support available through ScalAgent Distributed Technologies
Open Messaging Queue	Sun Microsystems / Oracle	4.5.1	CDDL or GPLv2	Support for OMQ is undefined after the take over of Sun by Oracle
Fusion Middleware	Oracle	12c	Commercial	No stand alone JMS Server
RabbitMQ	SpringSource	2.7.0	Mozilla Public License	Mirroring queues for full active/active clustering. Commercial support by SpringSource
SonicMQ	Progress Software	8.5	Commercial	Real time replication without transactional rollback
SwiftMQ	ITT Software	8.1.0	Commercial	No ESB capability
WebSphere MQ	IBM	7.1	Commercial	Good Redbooks on the subject

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

JMS High Level Analysis Result

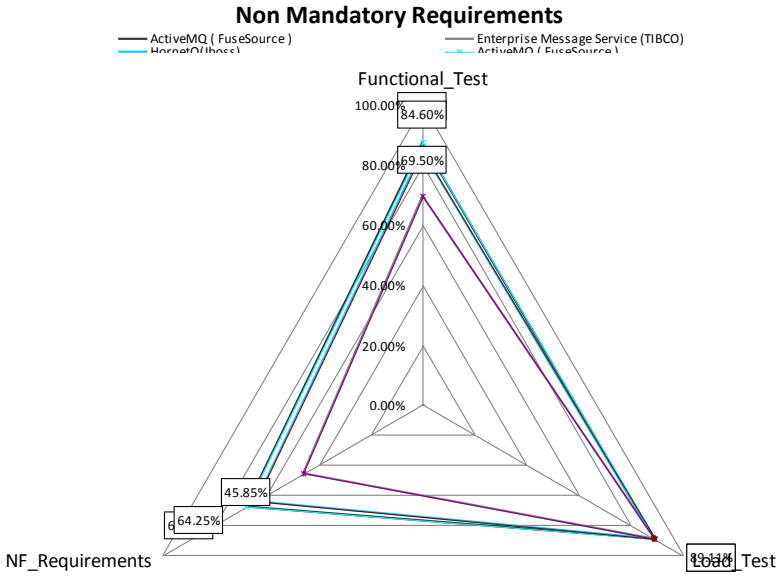


	Product		LifeCycle		Environment		Functionality		Specific			Overall
	Points	Percentage	Points	Percentage	Points	Percentage	Points	Percentage	Points	Percentage	Failed Criterion	Points
Messaging Server	**2410**		**880**		**1760**		**970**		**1570**		**0**	**7590**
ActiveMQ (Fusesource et. Al)	2082	86.39%	710	80.68%	1204	68.41%	934	96.29%	1080	68.79%	3	6010
Enterprise Message Service (TIBCO)	1750	72.61%	605	68.75%	1074	61.02%	899	92.68%	1070	68.15%	5	5398
FioranoMQ (Fiorano Software)	1238	51.37%	495	56.25%	1079	61.31%	950	97.94%	1060	67.52%	7	**4822**
HornetQ(Jboss)	2150	89.21%	790	89.77%	1200	68.18%	954	98.35%	920	58.60%	5	6014
JORAM (OW2 Consortium)	1598	66.31%	670	76.14%	670	38.07%	420	43.30%	1055	67.20%	17	**4413**
Open Messaging Queue (Sun)	876	36.35%	467	53.07%	905	51.42%	950	97.94%	1007	64.14%	15	**4205**
Fusion Middleware (Oracle)	1512	62.74%	590	67.05%	974	55.34%	860	88.66%	980	62.42%	7	**4916**
RabbitMQ (SpringSource)	1818	75.44%	720	81.82%	785	44.60%	950	97.94%	1126	71.72%	7	**5399**
SonicMQ (Progress Software)	1712	71.04%	605	68.75%	860	48.86%	970	100.00%	1010	64.33%	8	**5157**
SwiftMQ (ITT Software)	1610	66.80%	725	82.39%	950	53.98%	950	97.94%	900	57.32%	8	**5135**
WebSphere MQ (IBM)	1992	82.66%	675	76.70%	470	26.70%	780	80.41%	0	0.00%	9	**3917**

Note 2 : Initial data for the analysis was obtained from online sources only.

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

JMS PoC Results



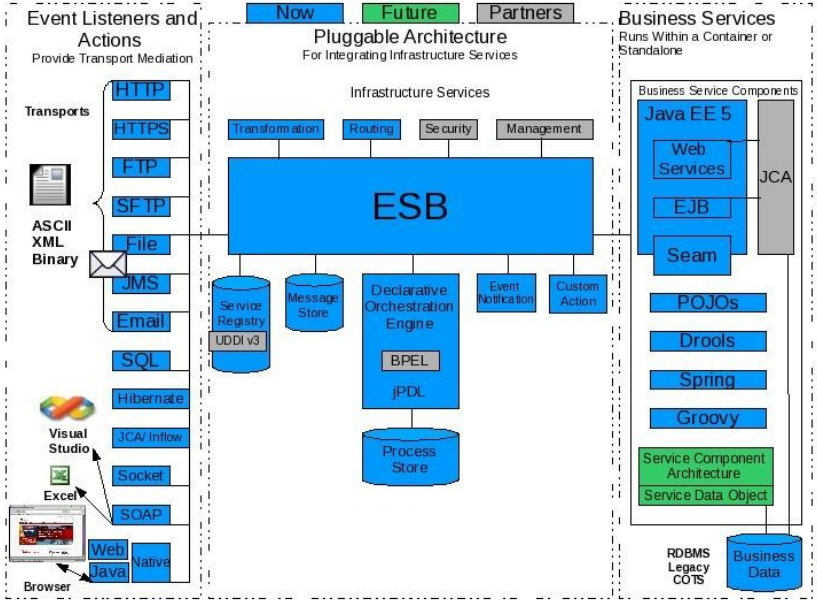
	Functional Tests		Load Test		NF Requirements		KO Criterias		Overall
	Points	Percentage	Points	Percentage	Points	Percentage	Failed Criterion	Passed Criterion	Points
Messaging	**1000**		**900**		**2000**		**0**	**68**	**3900**
ActiveMQ (FuseSource)	876	87.60%	802	89.11%	1354	67.70%	0	61	3032
Enterprise Message Service (T	695	69.50%	802	89.11%	917	45.85%	7	53	2414
HornetQ(Jboss)	846	84.60%	802	89.11%	1285	64.25%	0	65	2933

Note 2: All PoCs were done using VMWare instances, Oracle JDK 6 and NFSv3 for Message store.

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Evaluated ESB Solutions

Red Hat – SOA-P 5.3



Tibco – ActiveMatrix Service Bus

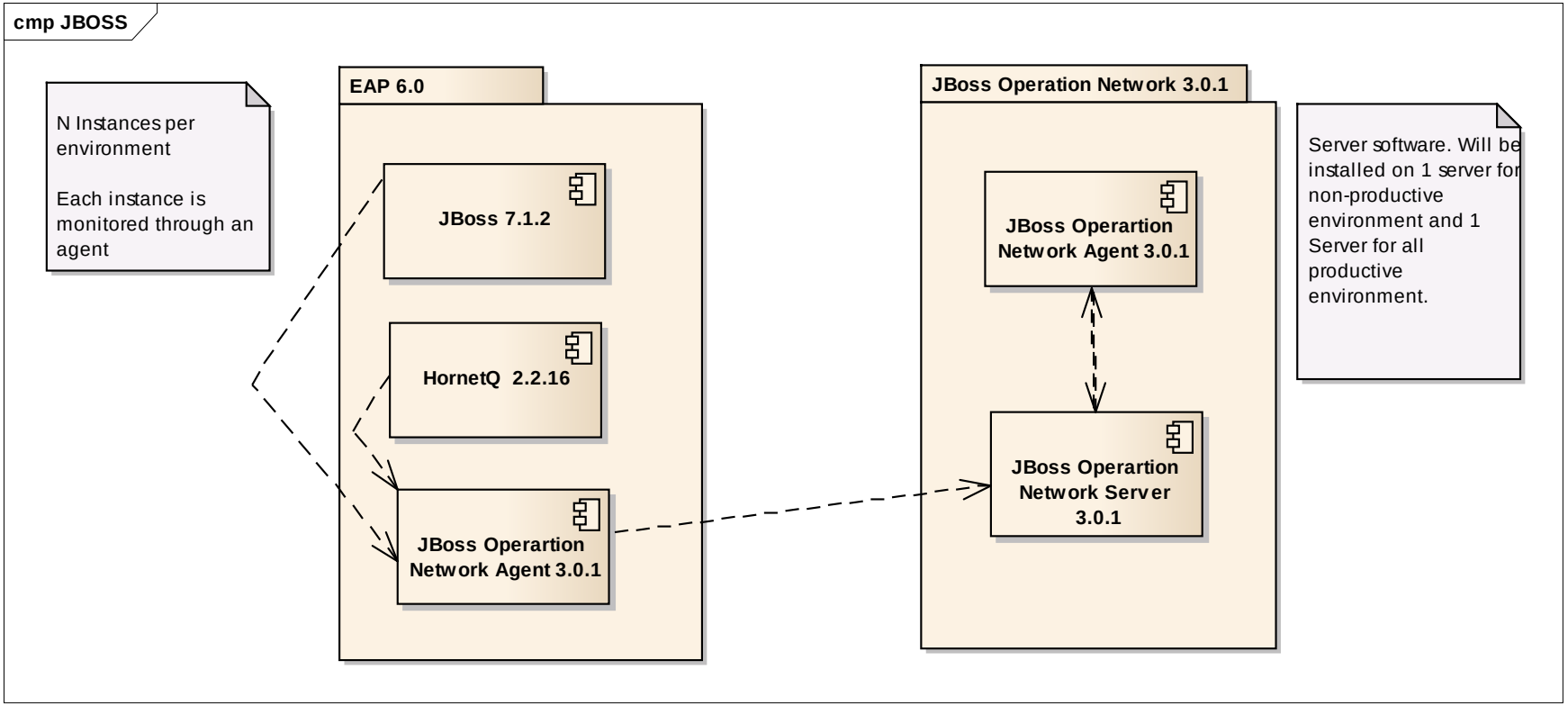
© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

ESB Requirements (Simplified)

Type	Weight	Non - KO criteria	KO criteria
Basic Functionality	700	12	5
Maintenance / Monitoring	150	8	4
Security	200	6	4
Failover (not transparent failover)	100	3	2
Automatism	100	7	3
Extendability	50	3	2
Development capabilities	250	9	2
Versioning	100	2	1
Costs	900	11	3
Quality	400	5	2
Service	500	7	2
HP Strategy	200	4	2
		77	**32**

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Architecture – EAP / SOA – P (Red Hat)



© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

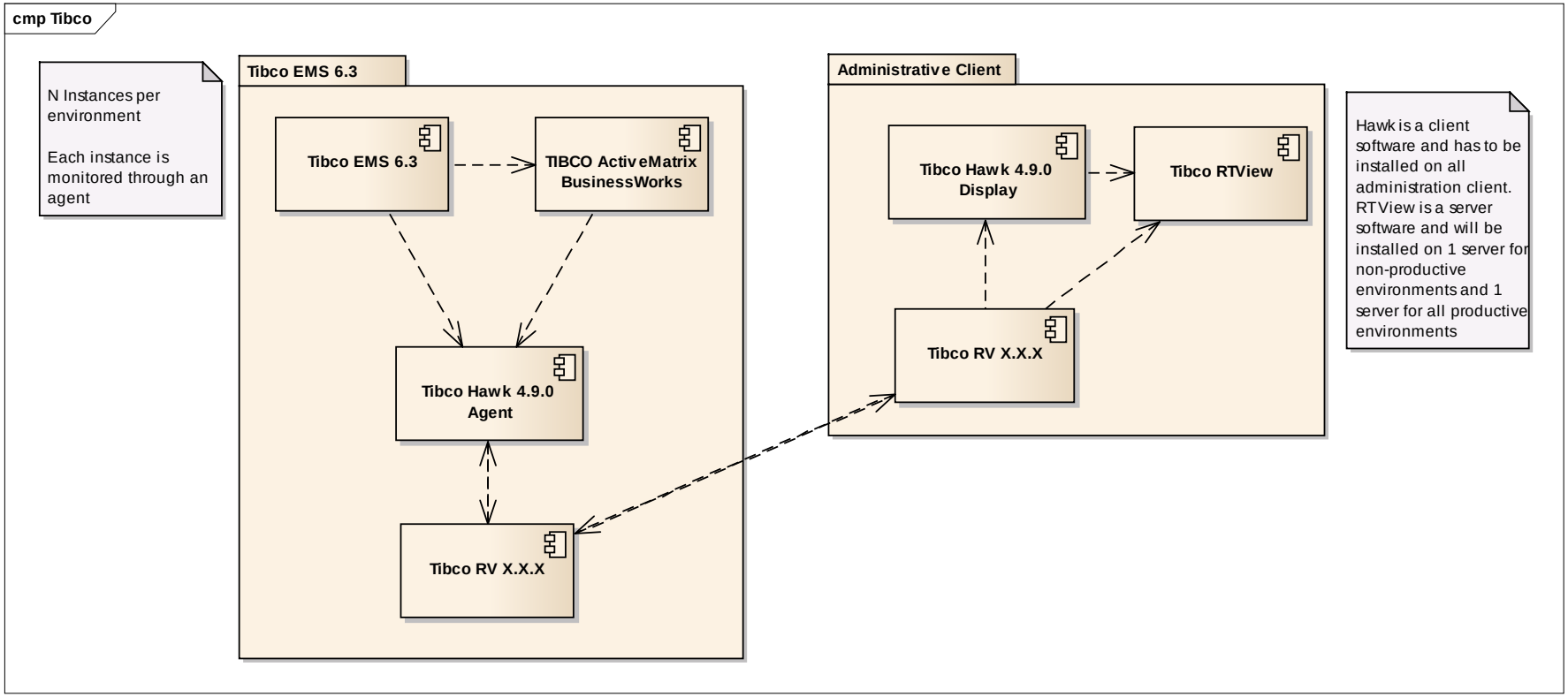
PoC – JBoss SOA-P 5.3 (Red Hat)

Strengths	Weaknesses
Very easy to install and maintain Very good Monitoring possibilities (JON) Lifecycle capabilities (JON) Future standard platform in HP Edge	No GUI for Flow design No integrated Debugging Documentation is not on “RedBook” (IBM) level No real customer references in CH / DE
Opportunities	**Threats**
JBoss + BPM are integrated in the whole RedHat platform Cloud ready Synergetic with other Red Hat products (JBoss AS)	Complete rewrite of the ESB in SOA-P 6.x Complete rewrite of the BPM in SOA-P 7 Loose coupling with other JBoss applications Acquisition Strategy unclear (FuseSource, Polymita)

Noteworthy Customers: HP, IBM, Intel, AMD, SAP, Dell, Cisco

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Architecture – ActiveMatrix (Tibco)



© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

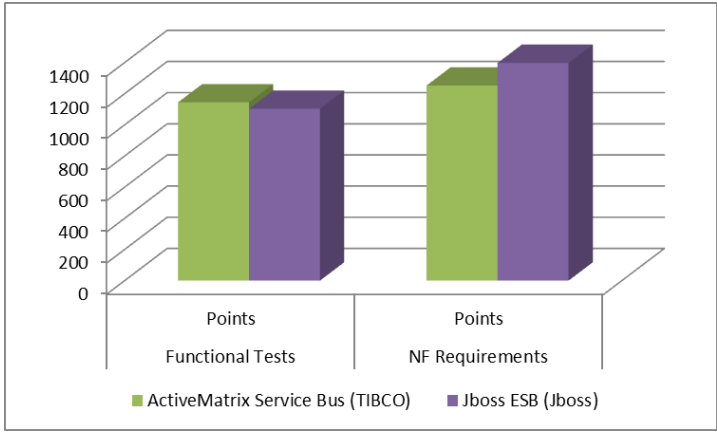
PoC – ActiveMatrix (Tibco)

Strenghts	Weaknesses
Fastest JMS Server overall GUI for Flow design Debugger integrated into solution Great support contacts	„Heavy“ Development installation (6 different products / installer) Development tools are completely proprietary and not java / JEE oriented Tibco EMS has requirement flaws (can be fixed)
Opportunities	**Threats**
BPM capabilities Huge available know-how pool @ HP HP Preferred Partner	Financial disadvantages (licenses) Tibco strategy is not JEE Complete separate development environment

Noteworthy Customers: Vodafone, Citibank Asia, Fed Ex, Pirelli, Delta Air Lines

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

ESB PoC Results

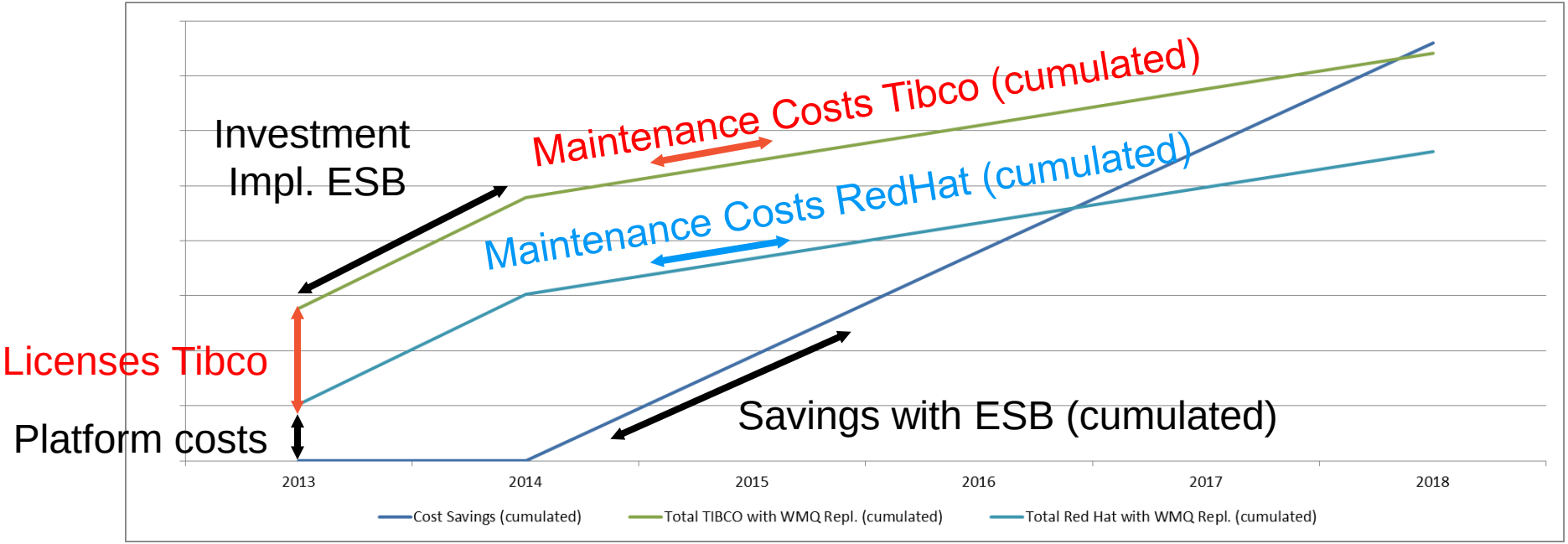


	Functional Tests		NF Requirements		KO Criterias			Overall
		Points		Points	Failed Criterion		Passed Criterion	Points
ESB	●	**1650**	●	**2000**	**0**	○	**68**	● **5450**
ActiveMatrix Service Bus (TIBC	◑	1143.75	◕	1252	0	◔	53	◔ 2395.75
Jboss ESB (Jboss)	◑	1102.5	●	1397	0	◔	59	◔ 2499.5

Note 2: All PoCs were done using VMWare instances, Oracle JDK 6 and NFSv3 for Message store.
Note 3: Not all tests could be performed due to the lack of availability of the key engineers and time constraint

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

Business Case - ESB



Assumption: Maintenance effort is the same for both RedHat SOA-P and Tibco ActiveMatrix

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

POC – Evaluation Summary

Technical view

-None of the ESB products failed any MUST requirements

-ActiveMatrix (TIBCO) has a better graphical layout and debugging features

-JBoss SOA-P (RED HAT) is better integrated in the current development landscape and in Java Enterprise

Business view

-JBoss SOA-P is financially better scalable (no license fees)

-ActiveMatrix has a big knowhow and capacity pool in the HP leveraged organization

Strategic view

-JBoss SOA-P (Red Hat) is easier to integrate

- Cost reduction WLS / WMQ migration
- Reselling Modules or processes without Royalties fee

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

PoC – Recommendation: RedHat SOA - P

- The initial costs between the two product was a deciding factor

- Redhat and Tibco both fulfill all priority 1 requirements

- Tibco EMS (Messaging) has more failed requirement

- Redhat is more align with the strategy of IBIS3G (Jboss Application Server, Red Hat Linux)

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

PoC – Lessons Learned

Commercial Software

Pros

- Very active sales department (might also have some very good management connections)
- Change to the software is possible if the use case is important
- Software has a bigger paying customer base

Cons

- Internal product decision was delayed because of said sales department
- Product palette might not be up to the newest standards (“we’ll make our own standards”)
- The fees for the product licenses are not transparent

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

PoC – Lessons Learned

Commercial OSS

Pros

- Technical support is very active and professional (that's where they generate revenue after all)
- Experts are available because of the community edition (see Eclipse, JBoss, etc)
- The quality of the commercially supported software is very high

Cons

- Documentation refers sometimes to the community open source edition (which is sometimes a little too "up-to-date" compared to the commercial OSS)
- Reference customer are published but it wasn't possible to visit one or have a conversation with IT architects
- Pre-Sale doesn't need to "woo" customer, the customer wants to go to them

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.

© Copyright 2012 Hewlett-Packard Development Company, L.P. The information contained herein is subject to change without notice.